Final Exam (Spring 2023)

ELEC70082 – Distributed Optimisation and Learning

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04/05/2023, 10:00 – 13:00

Name: _____

Read all of the following information before starting the exam:

- The exam is open book. Feel free to use any written or printed material you have brought. You can use and refer to any results from the lecture notes that might be useful in answering the questions.
- Calculators or other electronic devices are not needed for this exam.
- There are a total of 4 problems that total to 100 points. There are a total of 5 pages.
- You have 3 hours (10:00 - 13:00) to complete the exam.

Question 1. The following are true/false questions. Full credit will only be given for correct answers with correct justification, but partial credit is possible.

(a) [5 points] (true/false) A collection of K agents always exhibits a K-fold improvement in steady-state performance when collaborating through the Consensus+Innovations algorithm.

Solution. [2 points] The statement is false.

[3 points] K-fold gain in performance is the ultimate goal in distributed learning, but requires a homogeneous network of agents. Compare for example relations (10.2) and (10.5) in the lecture notes.

(b) [5 points] (true/false) Consider a network of K agents described by an adjacency matrix $A \in \mathbb{R}^{K \times K}$. Suppose the matrix is symmetric and doubly-stochastic, i.e., $A\mathbb{1} = \mathbb{1}$ and $A = A^{\mathsf{T}}$. It then follows that $\lim_{i \to \infty} A^i$ exists and is a rank-one matrix.

Solution. [2 points] The statement is false.

[3 points] This statement is the Perron-Frobenius theorem (relation (6.7) in the lecture notes), but is missing the strong-connectivity condition. Without this condition, it does not hold. One counterexample is:

$$A = I_M \Longrightarrow A^i = I_M \tag{1}$$

(c) [5 points] (true/false) Suppose a network of highly heterogeneous agents is sparsely connected. For moderate step-sizes and gradient approximation accuracies, it can then be expected that primal-dual algorithms for decentralized optimization will outperform penalty-based algorithms.

Solution. [2 points] The statement is true.

[3 points] The additional bias encountered by penalty-based algorithms scales with $\frac{1}{1-\lambda_2}$, which is large for sparse networks and $\|\nabla J_k(w^o)\|^2$, which is large for heterogeneous networks. So long as step-sizes are not too small, or gradient approximations are not too innaccurate, it will then hold that primal-dual algorithms outperform penalty-based ones. These trade-offs are covered in detail in Chapter 10.

(d) [5 points] (true/false) All else being equal, the Consensus+Innovations algorithm implemented over a more densely connected network will perform better than over a sparse network.

Solution. [2 points] The statement is true.

[3 points] The steady-state error of the Consensus+Innovations algorithm depends on the connectivity of the underlying network through $\frac{1}{1-\lambda_2}$, which is small for densely connected networks. See for example Theorem 9.1.

Question 2. Each agent in a network has access to a local objective $J_k(w)$ with $w \in \mathbb{R}^M$. At each iteration, each agent picks a random coordinate $\boldsymbol{m}_{k,i}$ by sampling uniformly from the set of integers between 1 and M. It then constructs local gradient approximations by differentiating only relative to the $\boldsymbol{m}_{k,i}$-th entry:

$$\widehat{\nabla J}_k(\boldsymbol{w}_{k,i-1}) \triangleq M \frac{\partial J_k(\boldsymbol{w}_{k,i-1})}{\partial w[\boldsymbol{m}_{k,i}]} e_{\boldsymbol{m}_{k,i}} \tag{2}$$

(a) [15 points] Show that this gradient approximation satisfies the gradient noise conditions:

$$\mathbb{E}\left\{\widehat{\nabla J}_k(\boldsymbol{w}_{k,i-1})|\boldsymbol{w}_{k,i-1}\right\} = \nabla J_k(\boldsymbol{w}_{k,i-1}) \tag{3}$$

$$\mathbb{E}\left\{\|\nabla J_k(\boldsymbol{w}_{k,i-1}) - \widehat{\nabla J}_k(\boldsymbol{w}_{k,i-1})\|^2|\boldsymbol{w}_{k,i-1}\right\} \leq \beta_k^2\|w_k^o - \boldsymbol{w}_{k,i-1}\|^2 + \sigma_k^2 \tag{4}$$

for suitably chosen constants β_k^2, σ_k^2.

Solution. **[5 points]** We begin by establishing the unbiasedness condition. To this end, we have:

$$\begin{aligned}
&\mathbb{E}\left\{\widehat{\nabla J}_k(\boldsymbol{w}_{k,i-1})|\boldsymbol{w}_{k,i-1}\right\} \\
&= \sum_{m=1}^{M} \mathbb{E}\left\{\widehat{\nabla J}_k(\boldsymbol{w}_{k,i-1})|\boldsymbol{w}_{k,i-1}, \boldsymbol{m}_{k,i} = m\right\} \cdot \Pr\{\boldsymbol{m}_{k,i} = m\} \\
&= \frac{1}{M}\sum_{m=1}^{M} M\frac{\partial J_k(\boldsymbol{w}_{k,i-1})}{\partial w[m]} e_m \\
&= \sum_{m=1}^{M} \frac{\partial J_k(\boldsymbol{w}_{k,i-1})}{\partial w[m]} e_m \\
&= \nabla J_k(\boldsymbol{w}_{k,i-1})
\end{aligned} \tag{5}$$

[10 points] For the variance, we have:

$$\begin{aligned}
&\mathbb{E}\left\{\left\|\nabla J_k(\boldsymbol{w}_{k,i-1}) - \widehat{\nabla J}_k(\boldsymbol{w}_{k,i-1})\right\|^2|\boldsymbol{w}_{k,i-1}\right\} \\
&= \sum_{m=1}^{M} \mathbb{E}\left\{\left(\frac{\partial J_k(\boldsymbol{w}_{k,i-1})}{\partial w[m]} - M\mathbb{1}_{k,i,m}\frac{\partial J_k(\boldsymbol{w}_{k,i-1})}{\partial w[m]}\right)^2|\boldsymbol{w}_{k,i-1}\right\} \\
&= \sum_{m=1}^{M} \mathbb{E}(1 - M\mathbb{1}_{k,i,m})^2\left(\frac{\partial J_k(\boldsymbol{w}_{k,i-1})}{\partial w[m]}\right)^2 \\
&= \sum_{m=1}^{M} \Big(\mathbb{E}\left\{(1 - M\mathbb{1}_{k,i,m})^2|\mathbb{1}_{k,i,m} = 1\right\} \cdot \Pr\{\mathbb{1}_{k,i,m} = 1\} \\
&\quad + \mathbb{E}\left\{(1 - M\mathbb{1}_{k,i,m})^2|\mathbb{1}_{k,i,m} = 0\right\} \cdot \Pr\{\mathbb{1}_{k,i,m} = 0\}\Big)\left(\frac{\partial J_k(\boldsymbol{w}_{k,i-1})}{\partial w[m]}\right)^2 \\
&= \sum_{m=1}^{M} \left(\frac{1}{M}(1-M)^2 + \left(1 - \frac{1}{M}\right)\right)\left(\frac{\partial J_k(\boldsymbol{w}_{k,i-1})}{\partial w[m]}\right)^2 \\
&= \sum_{m=1}^{M} \left(\frac{1}{M} - 2 + M + 1 - \frac{1}{M}\right)\left(\frac{\partial J_k(\boldsymbol{w}_{k,i-1})}{\partial w[m]}\right)^2
\end{aligned}$$

$$
\begin{aligned}
&= (M-1)\sum_{m=1}^{M}\left(\frac{\partial J_k(\boldsymbol{w}_{k,i-1})}{\partial w[m]}\right)^2 \\
&= (M-1)\,\|\nabla J_k(\boldsymbol{w}_{k,i-1})\|^2 \\
&= (M-1)\,\|\nabla J_k(w_k^o) - \nabla J_k(\boldsymbol{w}_{k,i-1})\|^2 \\
&= (M-1)\,\delta_k^2\|w_k^o - \boldsymbol{w}_{k,i-1}\|^2
\end{aligned} \tag{6}
$$

(b) [10 points] Write down the Exact Diffusion recursions for this particular choice of gradient approximation. How do you expect the limiting error $\lim_{i\to\infty}\mathbb{E}\|w^o - \boldsymbol{w}_{k,i}\|^2$ to compare to the implementation where you use true gradients $\nabla J_k(w)$ at each agent?

Solution. [5 points] We have from (8.65)–(8.67):

$$
\boldsymbol{\psi}_{k,i} = \boldsymbol{w}_{k,i-1} - \mu M\frac{\partial J_k(\boldsymbol{w}_{k,i-1})}{\partial w[\boldsymbol{m}_{k,i}]}e_{\boldsymbol{m}_{k,i}} \tag{7}
$$

$$
\boldsymbol{\phi}_{k,i} = \boldsymbol{\psi}_{k,i} + \boldsymbol{w}_{k,i-1} - \boldsymbol{\psi}_{k,i-1} \tag{8}
$$

$$
\boldsymbol{w}_{k,i} = \sum_{\ell\in\mathcal{N}_k} a_{\ell k}\boldsymbol{\phi}_{\ell,i-1} \tag{9}
$$

[5 points] Since $\sigma_k^2 = 0$, we expect this algorithm to not exhibit any steady-state error. As such, the performance of the coordinate descent based algorithm, in steady-state, can be expected to match the solution employing full gradients.

Question 3. We consider a network of K agents, each with local objective:

$$J_k(w) = \frac{1}{2}\mathbb{E}\Big(\boldsymbol{\gamma}_k - \boldsymbol{h}_k^{\mathsf{T}} w\Big)^2 \tag{10}$$

The agents employ the diffusion algorithm with locally constructed mini-batch approximations of the gradient:

$$\boldsymbol{\psi}_{k,i} = \boldsymbol{w}_{k,i-1} + \frac{\mu}{B}\sum_{b=1}^{B} \boldsymbol{h}_{k,i,b}\left(\boldsymbol{\gamma}_{k,i,b} - \boldsymbol{h}_{k,i,b}^{\mathsf{T}}\boldsymbol{w}_{k,i-1}\right) \tag{11}$$

$$\boldsymbol{w}_{k,i} = \sum_{\ell\in\mathcal{N}_k} a_{\ell k}\boldsymbol{\psi}_{\ell,i} \tag{12}$$

(a) [10 points] Determine an expression for the limiting performance $\lim_{i\to\infty}\mathbb{E}\|w^o - \boldsymbol{w}_{k,i}\|^2$ in terms of algorithm parameters and data statistics. You can assume the step-size to be small enough, so that any higher-order terms (i.e., $O(\mu^2)$) can be disregarded, and that all agents are homogenous (i.e., their data is i.i.d.).

Solution. [10 points] The limiting performance of the diffusion algorithm in homogenous environments is given by $\mu\frac{\sigma^2}{K\nu}$. For the least-squares problem with mini-batch gradient approximations we have $\sigma^2 = \frac{\sigma_v^2\mathrm{Tr}(R_h)}{B}$ and hence:

$$\lim_{i\to\infty}\mathbb{E}\|w^o - \boldsymbol{w}_{k,i}\|^2 = \frac{\mu\sigma_v^2\mathrm{Tr}\left(R_h\right)}{KB\lambda_{\min}(R_h)} \tag{13}$$

Moving forward, agents are faced with communication constraints, and hence exchange quantized estimates:

$$\boldsymbol{\psi}_{k,i} = \boldsymbol{w}_{k,i-1} + \frac{\mu}{B}\sum_{b=1}^{B} \boldsymbol{h}_{k,i,b}\left(\boldsymbol{\gamma}_{k,i,b} - \boldsymbol{h}_{k,i,b}^{\mathsf{T}}\boldsymbol{w}_{k,i-1}\right) \tag{14}$$

$$\boldsymbol{\phi}_{k,i} = \boldsymbol{\mathcal{Q}}_k(\boldsymbol{\psi}_{k,i}) \tag{15}$$

$$\boldsymbol{w}_{k,i} = \sum_{\ell\in\mathcal{N}_k} a_{\ell k}\boldsymbol{\phi}_{\ell,i} \tag{16}$$

Where the quantizer satisfies the typical conditions:

$$\mathbb{E}\left\{\boldsymbol{\mathcal{Q}}_k(\boldsymbol{\psi}_{k,i})|\boldsymbol{\psi}_{k,i}\right\} = \boldsymbol{\psi}_{k,i} \tag{17}$$

$$\mathbb{E}\left\{\left\|\boldsymbol{\psi}_{k,i} - \boldsymbol{\mathcal{Q}}_k(\boldsymbol{\psi}_{k,i})\right\|^2|\boldsymbol{\psi}_{k,i}\right\} \le \beta_q^2\|\boldsymbol{\psi}_{k,i}\|^2 + \sigma_q^2 \tag{18}$$

with common parameters β_q^2, σ_q^2.

(b) [10 points] Determine an expression for the limiting performance $\lim_{i\to\infty}\mathbb{E}\|w^o - \boldsymbol{w}_{k,i}\|^2$ in terms of algorithm parameters, data statistics and quantizer parameters.

Solution. [10 points] From (13.30) we have:

$$\lim_{i\to\infty}\mathbb{E}\|w^o - \boldsymbol{w}_{k,i}\|^2 = O\left(\frac{\mu\sigma_v^2\mathrm{Tr}\left(R_h\right)}{KB\lambda_{\min}(R_h)}\right) + O\left(\frac{\mu^{-1}\overline{\sigma}_q^2}{K\lambda_{\min}(R_h)}\right) \tag{19}$$

where $\overline{\sigma}_q^2 = 2\beta_k^2\|w^o\|^2 + \sigma_q^2$.

(c) [10 points] Suppose each agent employs the same quantization scheme with T bits, and the quantization parameters scale as:

$$\beta_q^2 = c_1 2^{-T} \tag{20}$$

$$\sigma_q^2 = c_2 2^{-T} \tag{21}$$

Determine a sufficient condition on μ and T to ensure:

$$\lim_{i\to\infty} \mathbb{E}\|w^o - \boldsymbol{w}_{k,i}\|^2 \leq \epsilon \tag{22}$$

Solution. [5 points] We split the two terms in the answer to part (b) and first ensure:

$$O\left(\frac{\mu\sigma_v^2 \mathrm{Tr}\left(R_h\right)}{KB\lambda_{\min}(R_h)}\right) \leq \frac{\epsilon}{2} \Longleftrightarrow \mu \leq O\left(\frac{\epsilon KB\lambda_{\min}(R_h)}{2\sigma_v^2 \mathrm{Tr}\left(R_h\right)}\right) \tag{23}$$

[5 points] Then:

$$O\left(\frac{\mu^{-1}\overline{\sigma}_q^2}{K\lambda_{\min}(R_h)}\right) \leq \frac{\epsilon}{2} \Longleftrightarrow \overline{\sigma}_q^2 \leq O\left(\frac{\mu\epsilon K\lambda_{\min}(R_h)}{2}\right) \tag{24}$$

To ensure this condition, we need:

$$2^{-T} \leq O\left(\mu\epsilon K\lambda_{\min}(R_h)\right) \Longleftrightarrow T \geq \log_2\left(\mu\epsilon K\lambda_{\min}(R_h)\right) \tag{25}$$

Question 4. We will examine a random variant of the static consensus averaging algorithm, where at any point in time, agents employ a random weight matrix:

$$\boldsymbol{w}_{k,i} = \sum_{\ell \in \mathcal{N}_k} \boldsymbol{a}_{\ell k,i} \boldsymbol{w}_{\ell,i-1} \tag{26}$$

Each agent is initialized at some deterministic local signal g_k, i.e., $\boldsymbol{w}_{k,0} = g_k$. The random combination weights are collected in a random matix $\boldsymbol{A}_i$ that satisfies almost surely:

$$\boldsymbol{A}_i = \boldsymbol{A}_i^{\mathsf{T}} \tag{27}$$

$$\boldsymbol{A}_i \mathbb{1} = \mathbb{1} \tag{28}$$

Furthermore, the weight matrices are identically and independently distributed over time with mean and variance:

$$\mathbb{E}\boldsymbol{A}_i = \overline{A} \tag{29}$$

$$\mathbb{E}\big\|\boldsymbol{A}_i - \overline{A}\big\|^2 = \sigma_A^2 \tag{30}$$

and we assume $\overline{A}$ to be a primitive matrix with mixing rate $\lambda_2 = \rho\left(\overline{A} - \frac{1}{K}\mathbb{1}\mathbb{1}^{\mathsf{T}}\right) < 1$.

(a) [10 points] Determine an expression for the evolution of the network centroid:

$$\boldsymbol{w}_{c,i} \triangleq \frac{1}{K}\sum_{k=1}^{K} \boldsymbol{w}_{k,i} \tag{31}$$

Solution. [10 points] Since $\boldsymbol{A}_i$ is doubly-stochastic almost surely, it holds that:

$$\boldsymbol{w}_{c,i} = \frac{1}{K}\sum_{k=1}^{K} \boldsymbol{w}_{k,i} = \frac{1}{K}\sum_{k=1}^{K}\sum_{\ell \in \mathcal{N}_k} \boldsymbol{a}_{\ell k,i}\boldsymbol{w}_{\ell,i-1} = \frac{1}{K}\sum_{\ell=1}^{K}\sum_{k=1}^{K} \boldsymbol{a}_{\ell k,i}\boldsymbol{w}_{\ell,i-1} = \boldsymbol{w}_{c,i-1} \tag{32}$$

(b) [15 points] Determine an expression for the evolution of mean estimates:

$$\overline{w}_{k,i} \triangleq \mathbb{E}\boldsymbol{w}_{k,i} \tag{33}$$

Do they converge? If so, what is their limiting point and rate of convergence?
Solution. [5 points] Taking expectations of the random consensus recursion, we have:

$$\overline{w}_{k,i} = \sum_{\ell \in \mathcal{N}_k} \overline{a}_{\ell k,i}\overline{w}_{\ell,i-1} \tag{34}$$

[10 points] We observe that the mean estimates follow a deterministic average consensus algorithm driven by the weight matrix $\overline{A}$. It then follows that the mean estimates will converge to $\frac{1}{K}\sum_{k=1}^{K} g_k$ at a rate determined by λ_2.